ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** $2\,Mg + O_2 \longrightarrow 2\,MgO$	Two Mg atoms combine with one O2 molecule; MgO is the formula of magnesium oxide.
2	**(a)** 10	Bases turn red litmus blue and have pH above 7.
3	**(b)** $Fe + CuSO_4$ solution	Fe is more reactive than Cu; others involve a less reactive metal.
4	**(b)** 4	O=C=O has two double bonds: 4 shared pairs.
5	**(d)** excretion	Kidneys filter blood and form urine.
6	**(a)** dendrite	Dendrite tip receives the signal.
7	**(a)** at a definite plane related to the whip-like flagellum	Leishmania has a definite orientation of division; Amoeba divides in any plane.
8	**(a)** Gregor Johann Mendel	Mendel worked on garden pea plants.
9	**(b)** second law of reflection	Statement of the second law.
10	**(c)** retina	Light-sensitive screen at the back.
11	**(b)** coulomb	1 C is the charge of about 6×10^{18} electrons.
12	**(d)** from the north pole to the south pole	By convention, outside the magnet.
13	**(c)** biotic and abiotic components	All living and non-living parts interact.
14	**(c)** $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2\,NaCl$	Ions are exchanged between the two compounds.
15	**(a)** 1.4	Stomach has hydrochloric acid.
16	**(c)** its easy loss of an electron	Sodium readily forms Na^+.
17	**(d)** long-chain carboxylic acids	Fatty acid salts.
18	**(d)** CO_2, water and energy	Glucose is fully oxidised.
19	**(b)** phototropism	Positive phototropism.
20	**(c)** pollen grain	Pollen grains carry the male gametes.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Mg is covered by a layer of magnesium oxide (and carbonate) (1); The layer is protective, so it must be rubbed off with sandpaper for burning (1).	1+1
22	Acids give H^+ ions only in the presence of water (1); Dry HCl has no H^+ ions free (1).	1+1
23	Mercury (1); Sodium (or potassium) (1).	1+1
24	Carbon has four valence electrons (1); Cannot easily lose or gain 4 electrons, so it shares them (1).	1+1
25	Autotrophs make their own food by photosynthesis (1); Heterotrophs depend on other organisms for food (1).	1+1
26	Reflex is automatic and involuntary; controlled by spinal cord (1); Walking is voluntary and controlled by the brain (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Iron is more reactive than copper and displaces it (1); Blue $CuSO_4$ changes to pale green $FeSO_4$ (1); $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ (1).	1+1+1
28	Acid + metal -> salt + hydrogen (1); $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1); Hydrogen burns with a pop sound (1).	1+1+1
29	Hammer: metals are malleable, non-metals break into pieces (1); Circuit: metals complete the circuit and the bulb glows (1); Non-metals (except graphite) do not conduct (1).	1+1+1
30	n-butane $CH_3CH_2CH_2CH_3$ (1); isobutane (2-methylpropane) $CH_3CH(CH_3)CH_3$ (1); Both are C_4H_{10} (1).	1+1+1
31	Energy is released (1); $C_6H_{12}O_6 + 6\,O_2 \longrightarrow 6\,CO_2 + 6\,H_2O + \text{energy}$ (1); Oxidation of glucose (1).	1+1+1
	OR: Receptor in the skin detects heat (1); Sensory neuron to spinal cord; relay neuron; motor neuron (1); Effector muscle withdraws the hand (1).	1+1+1

32	A bud develops as an outgrowth due to repeated cell division (1); The bud grows into a tiny individual with its own mouth and tentacles (1); It detaches when mature and becomes an independent Hydra (1).	1+1+1
	OR: F1: genotype Tt, phenotype tall (1); F2 phenotypic ratio 3 tall : 1 dwarf (1); F2 genotypic ratio 1 TT : 2 Tt : 1 tt (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Equation with equal atoms of each element on both sides (1); Fe3O4 contains 3 Fe, so put 3 before Fe (1); 4 O atoms in Fe3O4 need 4 H2O (1); Put 4 before H2 to balance hydrogen (1); $3\,Fe\,(s) + 4\,H_2O\,(g) \longrightarrow Fe_3O_4\,(s) + 4\,H_2\,(g)$ (1).	1+1+1+1+1
34	Bubbles on the surface of zinc (1); $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1); Gas burns with a pop sound near a flame (1); Test tube becomes warm (exothermic) (1); Acid + metal -> salt + hydrogen (1).	1+1+1+1+1
35	$4\,Na + O_2 \longrightarrow 2\,Na_2O$ (1); $2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); $4\,Al + 3\,O_2 \longrightarrow 2\,Al_2O_3$ (amphoteric) (1); $2\,Cu + O_2 \longrightarrow 2\,CuO$ (black) (1); Amphoteric oxides react with both acid and base (1).	1+1+1+1+1
36	Carbon has 4 valence electrons (1); Forms covalent bonds by sharing (1); Catenation forms chains and rings (1); Tetravalency: bonds with 4 atoms (1); CH_4: C shares 4 pairs with 4 H (1).	1+1+1+1+1
	OR: Chlorophyll absorbs light energy (1); Water splits into hydrogen and oxygen (1); CO_2 is reduced to carbohydrates (1); Desert plants take CO_2 at night and store it as an acid (1); Use it during the day for photosynthesis (1).	1+1+1+1+1
37	Dendrites receive information (1); Cell body (1); Axon carries impulse away (1); Nerve endings release chemical (1); Synapse transmits to next neuron (1).	1+1+1+1+1
	OR: Fission (binary and multiple): Amoeba, Plasmodium (1); Fragmentation: Spirogyra (1); Regeneration: Planaria (1); Budding: Hydra, yeast (1); Spore formation: Rhizopus (1).	1+1+1+1+1
38	Parents: TT x tt (1); F1: all Tt, tall (1); F1 self-pollinated: gametes T and t (1); F2: TT, Tt, Tt, tt (1); Phenotypic ratio 3 : 1 and genotypic ratio 1 : 2 : 1 (1).	1+1+1+1+1
	OR: At infinity: at F, real, point-sized (1); Beyond C: between F and C, real, inverted, diminished (1); At C: at C, real, inverted, same size (1); Between C and F: beyond C, real, inverted, enlarged; at F: at infinity, enlarged (1); Between F and P: behind the mirror, virtual, erect, enlarged (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** all round yellow	Round and yellow are dominant.
2	**(d)** in vacuum to that in the medium	n = c/v.
3	**(d)** a concave lens	The concave lens diverges the rays.
4	**(a)** directly proportional to the potential difference, at constant temperature	V = IR.
5	**(d)** a bar magnet	One end behaves as N and the other as S.
6	**(a)** producers	Green plants fix energy.
7	**(d)** zinc is more reactive and displaces copper from its salt	Reactive metal displaces less reactive metal.
8	**(a)** magnesium hydroxide	$Mg(OH)_2$ neutralises acid.
9	**(b)** electrolysis of molten chlorides	They are too reactive for carbon reduction.
10	**(c)** catenation	Catenation.
11	**(c)** nephron	Each kidney has about a million nephrons.
12	**(b)** adrenaline	Adrenal glands.
13	**(a)** testes	Testes are the male sex organs.
14	**(c)** the type of sperm that fertilises the egg	Father's sperm may carry X or Y.
15	**(b)** a converging lens	It converges parallel rays.
16	**(c)** dispersion	Newton's prism experiment.
17	**(d)** R1 + R2 + R3	Rs = R1 + R2 + R3.
18	**(b)** induced current in a coil due to a changing magnetic field	It was discovered by Faraday.
19	**(c)** stratosphere	In the upper atmosphere.
20	**(d)** Displacement	More reactive Al displaces Fe from its oxide.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Variations help some individuals survive when the environment changes (1); This ensures the survival of the species over generations (1).	1+1
22	Heredity: transmission of characters from parents to offspring (1); Variation: differences among individuals of a species (1).	1+1
23	Angle of incidence equals the angle of reflection (1); Incident ray, reflected ray and normal lie in the same plane (1).	1+1
24	Ability of the eye lens to adjust its focal length (1); To see both near and distant objects clearly (1).	1+1
25	Current when 1 coulomb of charge flows through a point in 1 second (2).	2
26	Right-hand thumb rule: hold the wire in the right hand with the thumb along the current (1); Curled fingers show the direction of the field (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Virtual and erect (1); Same size as the object (1); Laterally inverted; as far behind as the object is in front (1).	1+1+1
28	Cornea: transparent front; most refraction (1); Iris and pupil control the light entering (1); Retina is the screen with rods and cones (1).	1+1+1
29	Q = I t = 0.5 x 600 = 300 C (1); n = Q / e (1); n = 300 / 1.6×10^{-19} = 1.875×10^{21} (1).	1+1+1
30	Lines emerge from N and merge at S outside the magnet (1); They form closed loops (1); Closer lines mean a stronger field and no two lines intersect (1).	1+1+1

31	Food chain: a single linear sequence of organisms (1); Food web: many interconnected food chains (1); Food web gives stability to the ecosystem (1).	1+1+1
	OR: Equation with equal number of atoms of each element on both sides (1); Balancing by hit and trial (1); $3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$ (1).	1+1+1
32	Electrolysis of brine (aqueous NaCl) (1); $2\,NaCl + 2\,H_2O \longrightarrow 2\,NaOH + Cl_2 + H_2$ (1); Products: NaOH, chlorine, hydrogen (1).	1+1+1
	OR: Calcination: $ZnCO_3 \longrightarrow ZnO + CO_2$ (1); Reduction: $ZnO + C \longrightarrow Zn + CO$ (1); Carbon acts as reducing agent (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Cornea: transparent window that refracts light (1); Iris: controls pupil size (1); Pupil: allows light to enter (1); Lens: focuses the image on the retina (1); Retina and optic nerve: convert light into electrical signals sent to the brain (1).	1+1+1+1+1
34	V is proportional to I at constant temperature (1); Connect cell, ammeter, resistor and key in series and a voltmeter across the resistor (1); Take readings for different numbers of cells (1); The V-I graph is a straight line through the origin (1); The slope of the graph gives the resistance (1).	1+1+1+1+1
35	Place a compass near a straight wire passing through a card (1); Switch on the current: the needle deflects (1); Reversing the current reverses the deflection (1); Iron filings form concentric circles (1); Right-hand thumb rule gives the direction (1).	1+1+1+1+1
36	Ecosystem: all living and non-living things interacting in an area (1); Biotic: producers, consumers, decomposers (1); Examples: plants, deer, fungi (1); Abiotic: air, water, soil, sunlight, temperature (1); Biotic components depend on abiotic components (1).	1+1+1+1+1
	OR: Oxidation: gain of oxygen/loss of hydrogen; reduction: loss of oxygen/gain of hydrogen (1); C is oxidised (1); ZnO is reduced (1); ZnO is the oxidising agent (1); C is the reducing agent (1).	1+1+1+1+1
37	Electrolysis of brine (1); $2\,NaCl + 2\,H_2O \longrightarrow 2\,NaOH + Cl_2 + H_2$ (1); Cl2 at anode, H2 at cathode (1); Uses: NaOH in soap; Cl_2 in water treatment (1); H_2 as fuel/ margarine (1).	1+1+1+1+1
	OR: Slow attack of air and moisture on metals (1); Rusting needs both air and water (1); Painting and oiling (1); Galvanisation and chrome plating (1); Alloying and anodising (1).	1+1+1+1+1
38	$CH_3COOH + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + H_2O$ (1); Conc. H_2SO_4 catalyst (1); Esters have fruity smell (1); Used in perfumes and flavours (1); $CH_3COOC_2H_5 + NaOH \longrightarrow CH_3COONa + C_2H_5OH$ (1).	1+1+1+1+1
	OR: Kidneys, ureters, bladder, urethra (1); Filtration in glomerulus (1); Reabsorption in tubules (1); Urine flows to bladder through ureter (1); Released through urethra (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** HCl	Egg-shell is $CaCO_3$; acids release CO_2.
2	**(a)** applying grease	Paint/zinc cannot be used on a pan used for cooking; grease is suitable.
3	**(d)** 16	3n + 1 = 16.
4	**(c)** transport of water	Xylem carries water and minerals.
5	**(d)** synapse	Chemical signal crosses it.
6	**(d)** Plasmodium	Plasmodium divides into many daughter cells at once.
7	**(c)** it has many contrasting characters and is easy to grow	Pea has clear contrasting traits and short life cycle.
8	**(d)** virtual, erect and of the same size as the object	Plane mirror image is virtual, erect, same size, laterally inverted.
9	**(b)** outer surface of the cornea	The cornea has the greatest change in refractive index.
10	**(a)** ampere	1 A = 1 C/s.
11	**(a)** the south pole to the north pole	They form closed loops.
12	**(b)** Temperature	Non-living factor.
13	**(c)** $BaSO_4$	Barium sulphate is insoluble in water.
14	**(d)** 5.5	Enamel dissolves below pH 5.5.
15	**(b)** calcium	Ca reacts less violently and floats.
16	**(b)** formation of micelles	Hydrophobic tails trap oil in micelles.
17	**(c)** lactic acid	Lactic acid causes cramps.
18	**(c)** gravity	Positive geotropism.
19	**(d)** ovary	After fertilisation the ovary ripens into fruit.
20	**(a)** 9 : 3 : 3 : 1	Four phenotypes appear.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Biotic: all living organisms (1); Abiotic: non-living factors like air, water, soil and temperature (1).	1+1
22	Paint forms a coating that keeps iron away from air and moisture (1); This prevents rusting (corrosion) (1).	1+1
23	Acid forms H^+ (hydronium) ions in water (1); Moving ions carry current (1).	1+1
24	Mixture of conc. HCl and conc. HNO_3 in 3 : 1 (1); It dissolves gold and platinum (1).	1+1
25	Atom or group that decides chemical properties (1); -OH (alcohol) (1).	1+1
26	Long and coiled, with villi (1); Villi increase surface area and are rich in blood vessels (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Ethanol is oxidised by alkaline $KMnO_4$ or acidified $K_2Cr_2O_7$ (1); $CH_3CH_2OH \longrightarrow CH_3COOH$ (1); Oxidising agent adds oxygen (1).	1+1+1
28	Filtration in glomerulus (1); Selective reabsorption of glucose, amino acids, water (1); Remaining waste forms urine (1).	1+1+1
29	Insulin: pancreas, sugar level (1); Thyroxine: thyroid, metabolism (1); Adrenaline: adrenal, emergency (1).	1+1+1
30	Anther produces pollen grains (1); Stigma receives pollen (1); Ovary contains ovules and becomes fruit (1).	1+1+1
31	Eggs always carry an X chromosome (1); Sperm can carry X or Y (1); The sperm that fertilises the egg decides the sex (1).	1+1+1
	OR: 1/v = 1/f + 1/u = 1/10 - 1/30 (1); v = +15 cm (1); m = v/u = -0.5 (1).	1+1+1
32	Ciliary muscles weaken (1); The eye lens loses flexibility (1); The near point recedes (presbyopia) (1).	1+1+1

	OR: Rs = 2 + 3 + 5 = 10 ohm (1); I = V / R (1); I = 0.6 A (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Light causes auxin to shift to the shaded side (1); Cells on that side elongate more (1); Shoot bends towards light (1); Auxin made at shoot tip (1); Experiment with a covered tip shows no bending (1).	1+1+1+1+1
34	A pair of ovaries produce eggs (1); The egg enters the oviduct (fallopian tube) (1); Fertilisation occurs in the oviduct (1); Zygote moves and is implanted in the uterus (1); Uterus opens into the vagina through the cervix (1).	1+1+1+1+1
35	He crossed plants having contrasting traits (1); F1 showed only one parent's trait (1); F2 had both traits in 3 : 1 (1); Traits are controlled by pairs of factors (1); Factors segregate in gametes and are independently inherited (1).	1+1+1+1+1
36	Power = 1/f in metres; unit dioptre (1); P1 = +4 D (1); P2 = -2 D (1); Net power = +2 D (1); Combination is converging with f = 50 cm (1).	1+1+1+1+1
	OR: Dispersion is splitting of white light (1); Light passes through a prism (1); Seven colour spectrum is formed (1); A second inverted prism recombines the colours (1); White light emerges, showing the prism does not create colours (1).	1+1+1+1+1
37	Work done to move charge Q across V is W = V Q (1); Q = I t (1); W = V I t (1); Using V = I R, W = I^2 R t (1); This energy is dissipated as heat H (1).	1+1+1+1+1
	OR: Mains supply of 220 V comes through live (red) and neutral (black) (1); Earth wire (green) is connected to the metal body (1); Fuse is placed in the live wire (1); Appliances are connected in parallel (1); Earth wire avoids shocks and fuse avoids overloading (1).	1+1+1+1+1
38	Increase in concentration of harmful chemicals along a food chain (1); Pesticide in water or crops (1); Consumed by herbivores and accumulates (1); Carnivores accumulate even more (1); Top consumers like humans have the highest level (1).	1+1+1+1+1
	OR: Burns with a dazzling white flame (1); $2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); Combination reaction (1); Product is a white powder, magnesium oxide (1); Its solution turns red litmus blue (basic) (1).	1+1+1+1+1

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